

OIBSIP Data Science Task 3 — Car Price Prediction

Objective

Predict the selling price of used cars using machine learning regression.

This notebook covers data loading, cleaning, EDA, feature engineering, preprocessing, model training, evaluation, comparison, findings, and conclusion.

Dataset file: `car_data.csv`

1. Import Libraries

Import the libraries needed for analysis, visualization, and machine learning.

```
In [7]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [8]: from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
from sklearn.pipeline import Pipeline

from sklearn.linear_model import LinearRegression
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor

from sklearn.metrics import (mean_absolute_error, mean_squared_error, r2_score)
```

```
In [9]: import warnings
warnings.filterwarnings("ignore")

print("Libraries imported successfully.")
```

Libraries imported successfully.

2. Load Dataset

The car dataset is loaded using Pandas.

```
In [13]: import pandas as pd

df = pd.read_csv(r"C:\Users\rrama_t5uaug6\Downloads\car_data.csv")

print("Dataset loaded successfully.")
df.head()
```

Dataset loaded successfully.

```
Out[13]:
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	Manual	0
1	sx4	2013	4.75	9.54	43000	Diesel	Dealer	Manual	0
2	ciaz	2017	7.25	9.85	6900	Petrol	Dealer	Manual	0
3	wagon r	2011	2.85	4.15	5200	Petrol	Dealer	Manual	0
4	swift	2014	4.60	6.87	42450	Diesel	Dealer	Manual	0

3. Dataset Overview

The dataset is inspected to understand its size, column names, data types, and overall structure.

```
In [15]: print("Dataset Shape:")
print(df.shape)

print("\nColumn Names:")
print(df.columns.tolist())

print("\nDataset Information:")
```

```
df.info()
```

Dataset Shape:
(301, 9)

Column Names:

```
['Car_Name', 'Year', 'Selling_Price', 'Present_Price', 'Kms_Driven', 'Fuel_Type', 'Seller_Type', 'Transmission', 'Owner']
```

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 301 entries, 0 to 300
```

```
Data columns (total 9 columns):
```

#	Column	Non-Null Count	Dtype
0	Car_Name	301 non-null	object
1	Year	301 non-null	int64
2	Selling_Price	301 non-null	float64
3	Present_Price	301 non-null	float64
4	Kms_Driven	301 non-null	int64
5	Fuel_Type	301 non-null	object
6	Seller_Type	301 non-null	object
7	Transmission	301 non-null	object
8	Owner	301 non-null	int64

```
dtypes: float64(2), int64(3), object(4)
```

```
memory usage: 21.3+ KB
```

4. First and Last Rows

The first and last records are displayed to verify that the dataset has been loaded correctly.

```
In [17]: display(df.head())
display(df.tail())
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	Manual	0
1	sx4	2013	4.75	9.54	43000	Diesel	Dealer	Manual	0
2	ciaz	2017	7.25	9.85	6900	Petrol	Dealer	Manual	0
3	wagon r	2011	2.85	4.15	5200	Petrol	Dealer	Manual	0
4	swift	2014	4.60	6.87	42450	Diesel	Dealer	Manual	0
	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner
296	city	2016	9.50	11.6	33988	Diesel	Dealer	Manual	0
297	brio	2015	4.00	5.9	60000	Petrol	Dealer	Manual	0
298	city	2009	3.35	11.0	87934	Petrol	Dealer	Manual	0
299	city	2017	11.50	12.5	9000	Diesel	Dealer	Manual	0
300	brio	2016	5.30	5.9	5464	Petrol	Dealer	Manual	0

5. Clean Column Names

Leading and trailing spaces are removed from column names to make the dataset easier to work with.

```
In [19]: df.columns = df.columns.str.strip()
print(df.columns.tolist())
```

```
['Car_Name', 'Year', 'Selling_Price', 'Present_Price', 'Kms_Driven', 'Fuel_Type', 'Seller_Type', 'Transmission', 'Owner']
```

6. Check for Missing Values

Missing values are checked before performing further analysis.

```
In [20]: missing_values = df.isnull().sum()

print("Missing values in each column:")
print(missing_values)

print("\nTotal missing values:")
print(missing_values.sum())
```

Missing values in each column:

```
Car_Name      0
Year          0
Selling_Price 0
Present_Price 0
Kms_Driven    0
Fuel_Type     0
Seller_Type   0
Transmission  0
Owner         0
dtype: int64
```

Total missing values:
0

7. Check for Duplicate Rows

Duplicate records are checked because repeated observations can affect the analysis and machine learning model.

```
In [21]: duplicate_count = df.duplicated().sum()

print("Number of duplicate rows:", duplicate_count)

if duplicate_count > 0:
    df = df.drop_duplicates().reset_index(drop=True)
    print("Duplicate rows removed.")
```

Number of duplicate rows: 2
Duplicate rows removed.

8. Descriptive Statistics

Descriptive statistics are used to understand the numerical variables in the dataset.

```
In [22]: df.describe(include="all")
```

```
Out[22]:
```

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission
count	299	299.000000	299.000000	299.000000	299.000000	299	299	299
unique	98	NaN	NaN	NaN	NaN	3	2	2
top	city	NaN	NaN	NaN	NaN	Petrol	Dealer	Manual
freq	26	NaN	NaN	NaN	NaN	239	193	260
mean	NaN	2013.615385	4.589632	7.541037	36916.752508	NaN	NaN	NaN
std	NaN	2.896868	4.984240	8.567887	39015.170352	NaN	NaN	NaN
min	NaN	2003.000000	0.100000	0.320000	500.000000	NaN	NaN	NaN
25%	NaN	2012.000000	0.850000	1.200000	15000.000000	NaN	NaN	NaN
50%	NaN	2014.000000	3.510000	6.100000	32000.000000	NaN	NaN	NaN
75%	NaN	2016.000000	6.000000	9.840000	48883.500000	NaN	NaN	NaN
max	NaN	2018.000000	35.000000	92.600000	500000.000000	NaN	NaN	NaN

9. Categorical Variable Analysis

The unique values of important categorical variables are inspected before they are encoded for machine learning.

```
In [23]: for column in ["Fuel_Type", "Seller_Type", "Transmission", "Owner"]:

    if column in df.columns:
        print(f"\n{column}:")
        print(df[column].value_counts())
```

```
Fuel_Type:
Fuel_Type
Petrol    239
Diesel    58
CNG        2
Name: count, dtype: int64
```

```
Seller_Type:
Seller_Type
Dealer    193
Individual 106
Name: count, dtype: int64
```

```
Transmission:
Transmission
Manual    260
Automatic  39
Name: count, dtype: int64
```

```
Owner:
Owner
0      288
1       10
3        1
Name: count, dtype: int64
```

10. Distribution of Selling Price

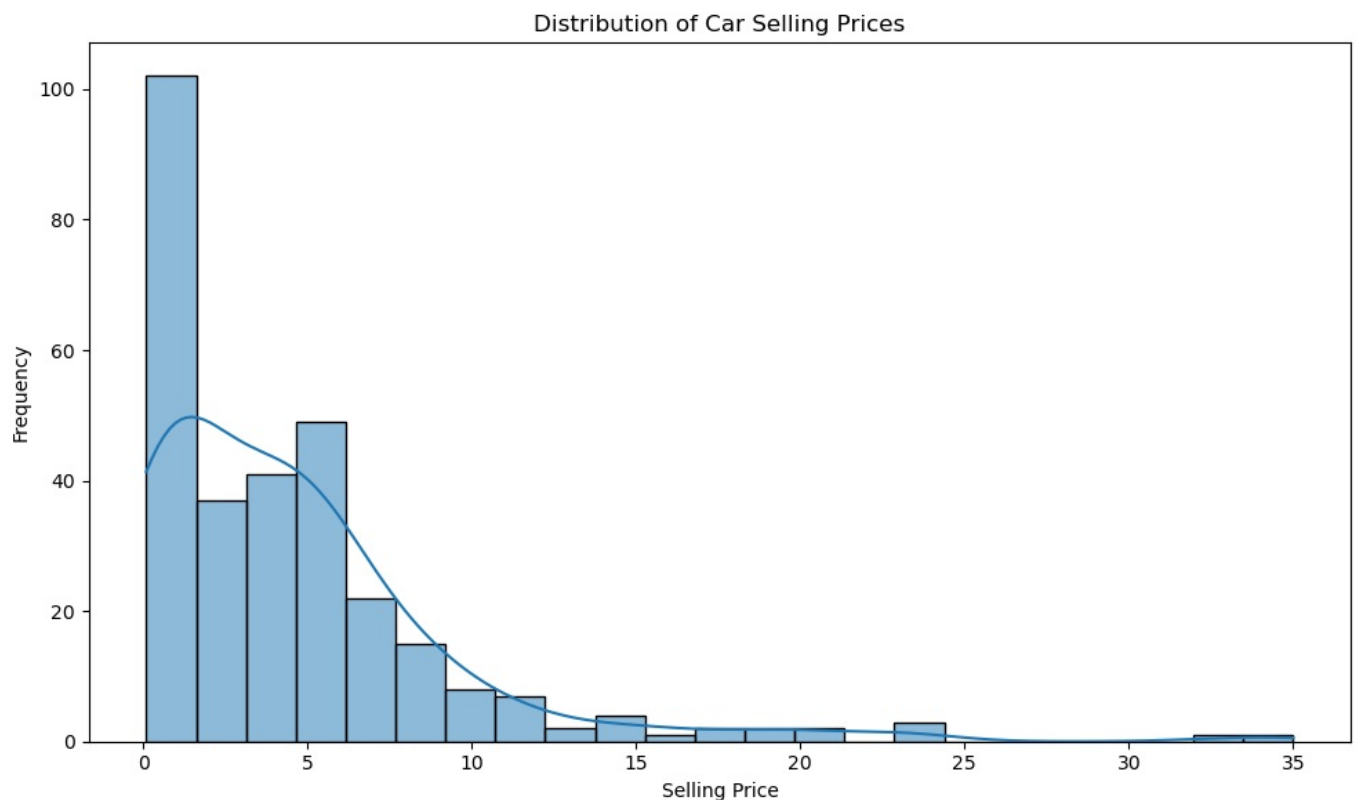
The distribution of selling prices is visualized to understand how prices are spread across the dataset.

```
In [24]: plt.figure(figsize=(10, 6))

sns.histplot(
    df["Selling_Price"],
    kde=True
)

plt.title("Distribution of Car Selling Prices")
plt.xlabel("Selling Price")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```



Observation

The selling price distribution shows that most cars in the dataset are concentrated in the lower price range, while a smaller number of cars have considerably higher selling prices. This indicates that the dataset is not evenly distributed.

across all price levels.

11. Present Price vs Selling Price

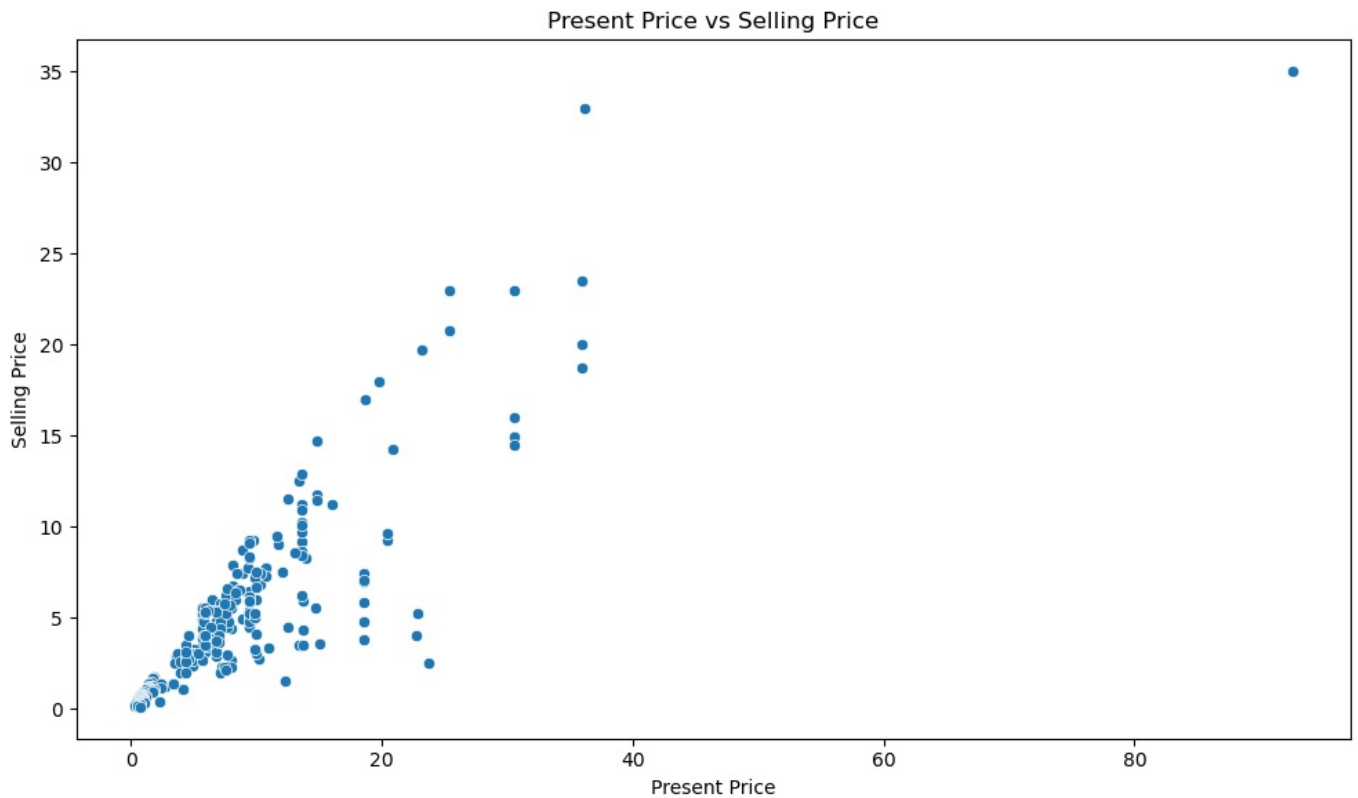
This analysis examines the relationship between the current market price and the actual selling price.

```
In [25]: plt.figure(figsize=(10, 6))

sns.scatterplot(data=df,
                x="Present_Price",
                y="Selling_Price")

plt.title("Present Price vs Selling Price")
plt.xlabel("Present Price")
plt.ylabel("Selling Price")

plt.tight_layout()
plt.show()
```



12. Year vs Selling Price

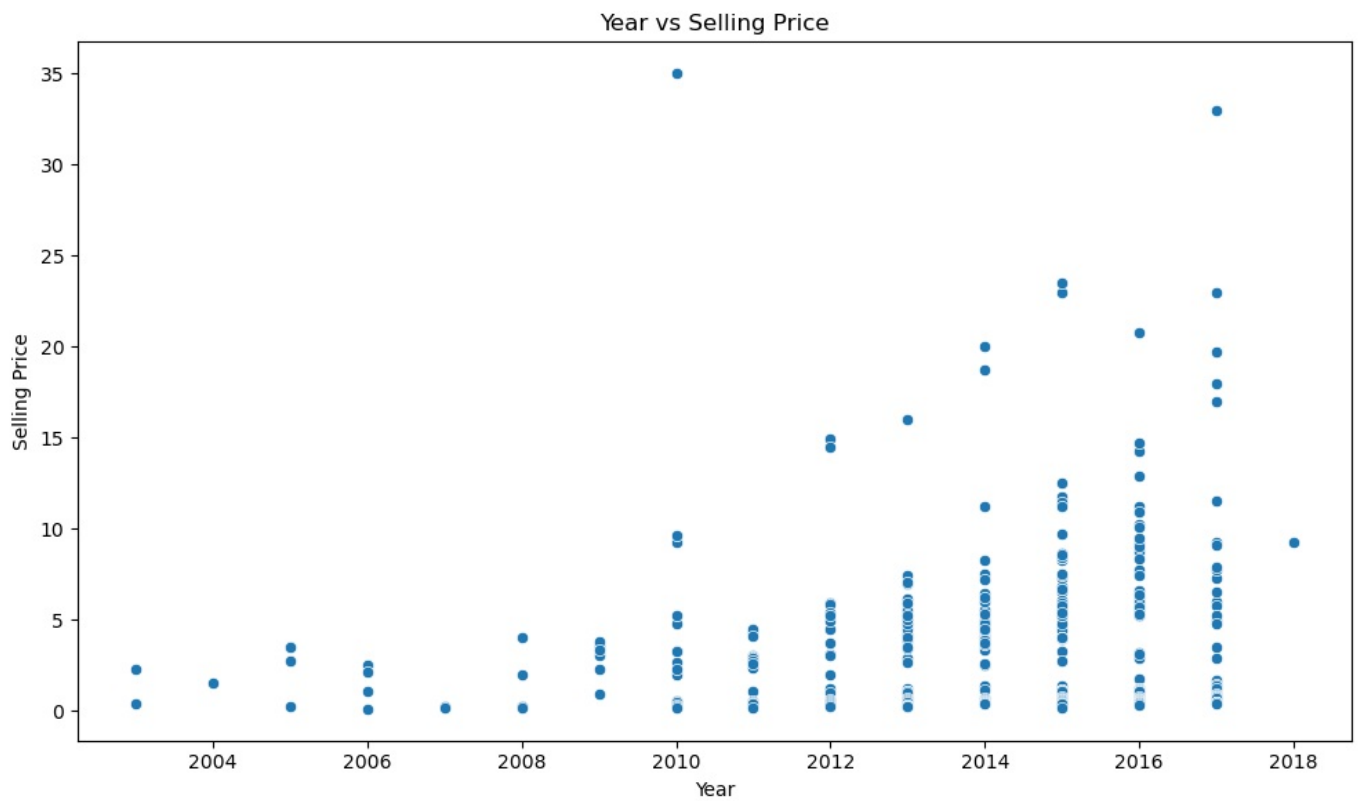
The relationship between the manufacturing year and selling price is examined.

```
In [27]: plt.figure(figsize=(10, 6))

sns.scatterplot(data=df,
                x="Year",
                y="Selling_Price")

plt.title("Year vs Selling Price")
plt.xlabel("Year")
plt.ylabel("Selling Price")

plt.tight_layout()
plt.show()
```



13. Kilometers Driven vs Selling Price

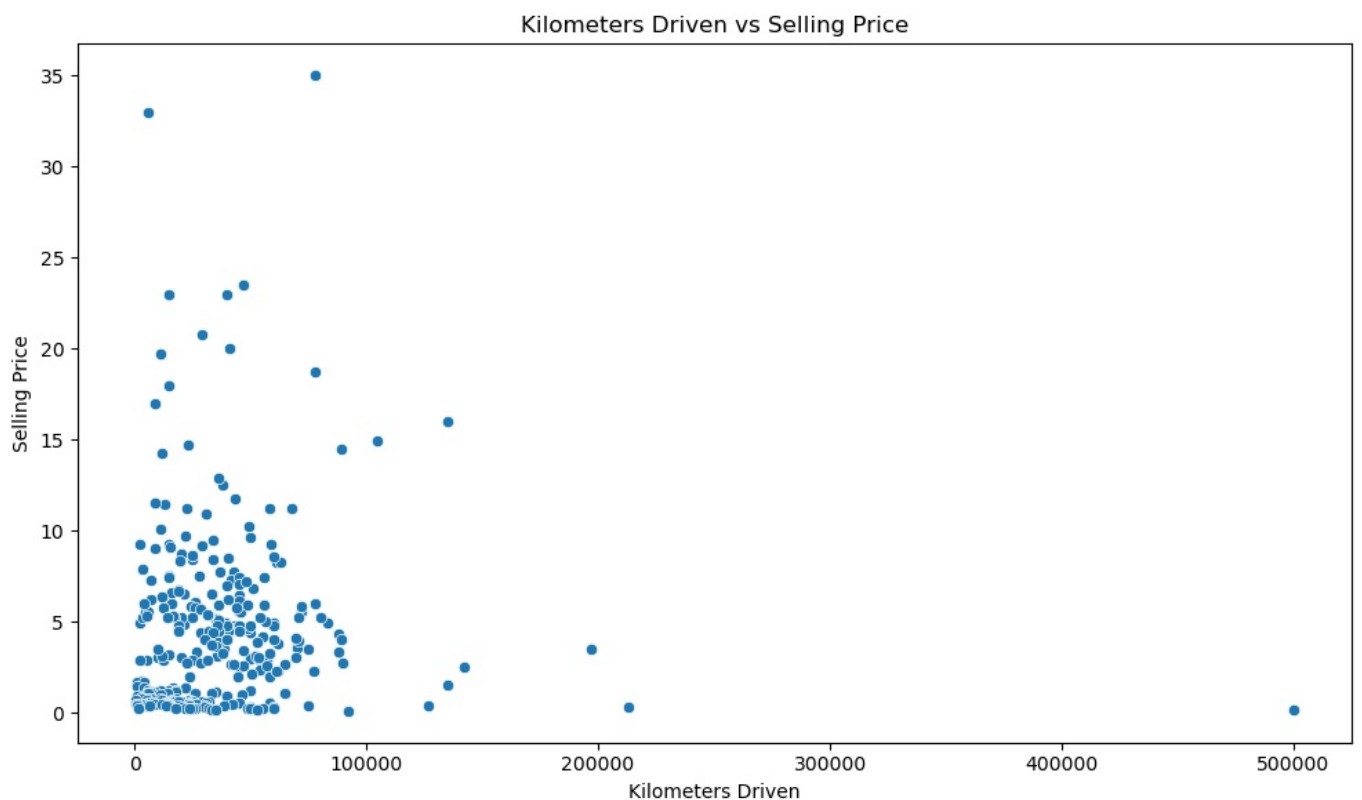
This analysis examines whether the number of kilometres driven is associated with the selling price.

```
In [28]: plt.figure(figsize=(10, 6))

sns.scatterplot(data=df,
                x="Kms_Driven",
                y="Selling_Price")

plt.title("Kilometers Driven vs Selling Price")
plt.xlabel("Kilometers Driven")
plt.ylabel("Selling Price")

plt.tight_layout()
plt.show()
```



14. Average Selling Price by Fuel Type

The average selling price is compared across different fuel types.

```
In [29]: fuel_summary = (
    df.groupby("Fuel_Type")["Selling_Price"]
    .mean()
    .sort_values(ascending=False))

print("Average selling price by fuel type:")
print(fuel_summary)

plt.figure(figsize=(8, 5))

fuel_summary.plot(kind="bar")

plt.title("Average Selling Price by Fuel Type")
plt.xlabel("Fuel Type")
plt.ylabel("Average Selling Price")

plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

Average selling price by fuel type:

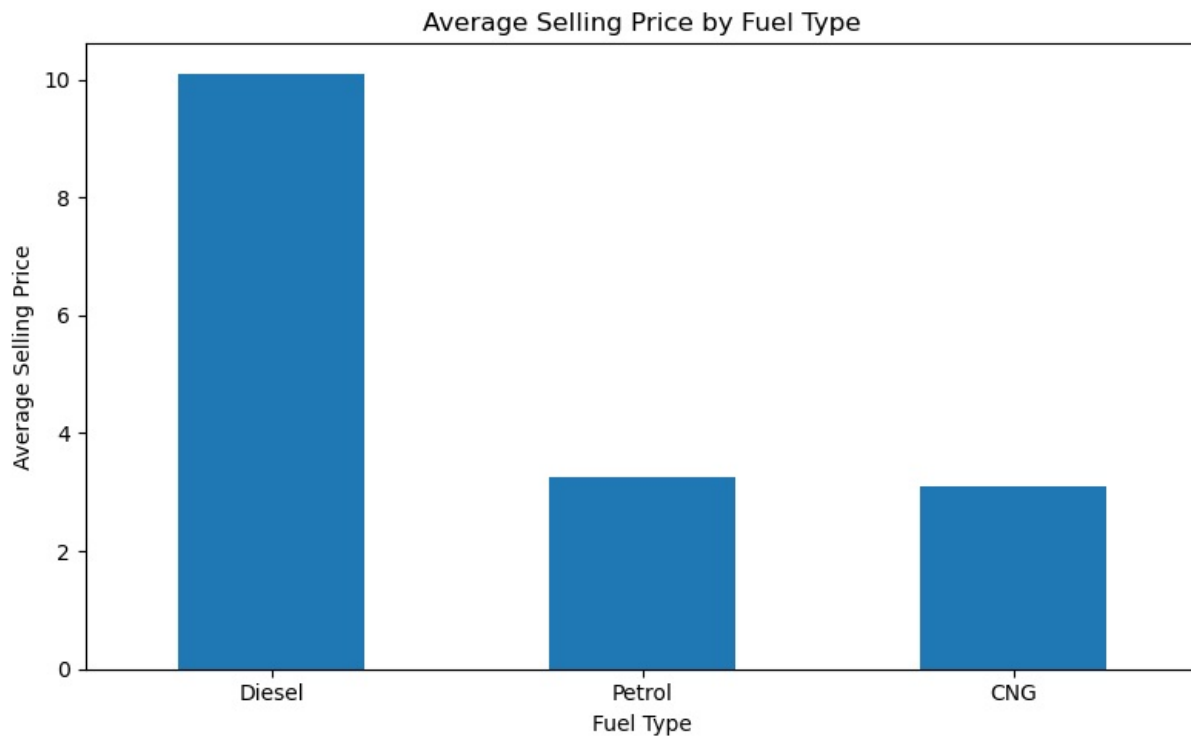
Fuel_Type

Diesel 10.102759

Petrol 3.264184

CNG 3.100000

Name: Selling_Price, dtype: float64



15. Selling Price by Fuel Type

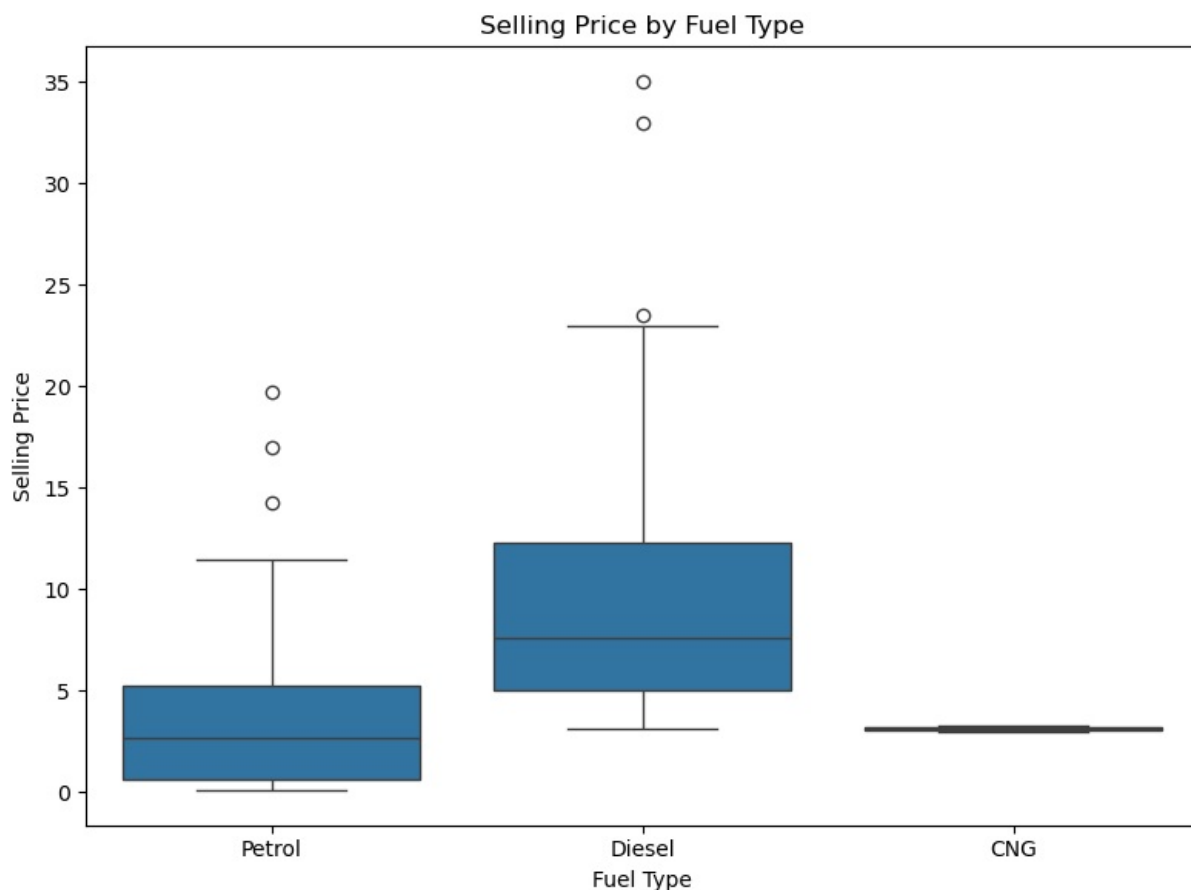
A box plot is used to compare the distribution of selling prices across different fuel types.

This helps show the spread and variation in selling prices rather than only comparing average values.

```
In [47]: plt.figure(figsize=(8, 6))

sns.boxplot(
    data=df,
    x="Fuel_Type",
    y="Selling_Price")

plt.title("Selling Price by Fuel Type")
plt.xlabel("Fuel Type")
plt.ylabel("Selling Price")
plt.tight_layout()
plt.show()
```



Observation

The box plot shows how selling prices vary across Petrol, Diesel, and CNG vehicles. Diesel vehicles show a wider range of selling prices in this dataset, while Petrol vehicles contain a larger number of observations.

16. Average Selling Price by Transmission

The average selling price is compared between different transmission types.

```
In [31]: transmission_summary = (
    df.groupby("Transmission")["Selling_Price"]
    .mean()
    .sort_values(ascending=False))

print("Average selling price by transmission:")
print(transmission_summary)

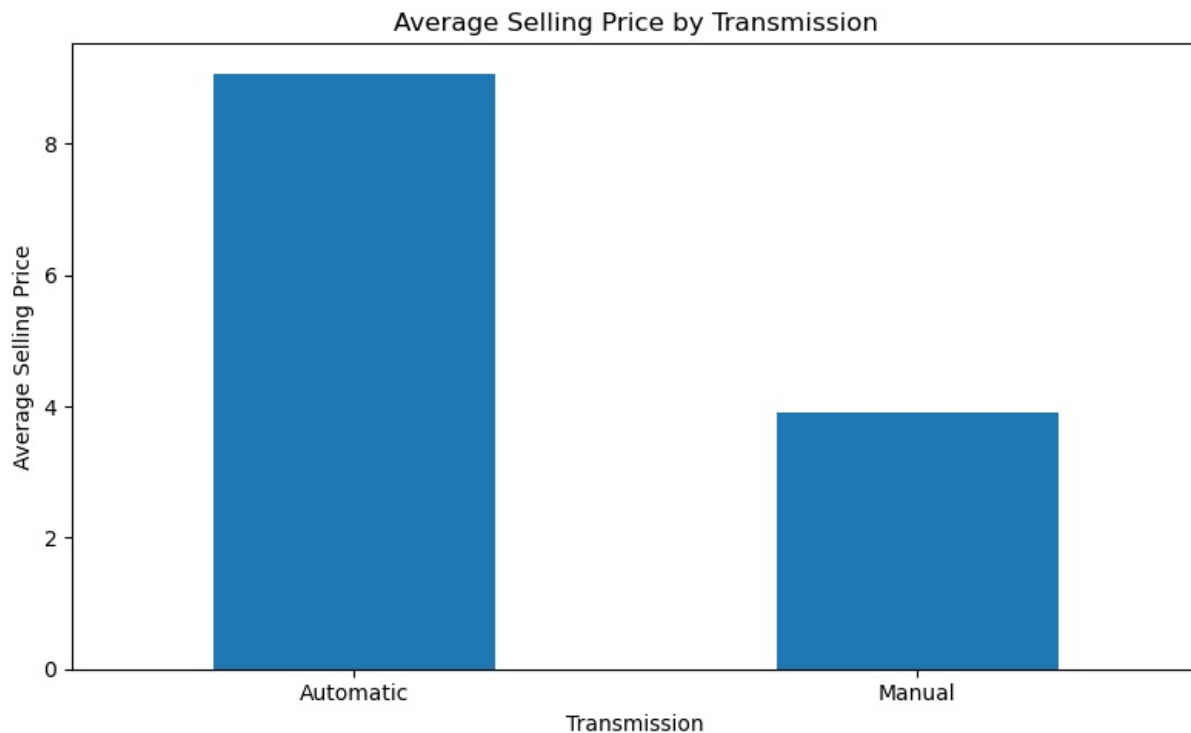
plt.figure(figsize=(8, 5))

transmission_summary.plot(kind="bar")

plt.title("Average Selling Price by Transmission")
plt.xlabel("Transmission")
plt.ylabel("Average Selling Price")

plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

```
Average selling price by transmission:
Transmission
Automatic    9.071795
Manual       3.917308
Name: Selling_Price, dtype: float64
```



17. Feature Engineering — Car Age

The manufacturing year is converted into a more useful feature called `Car_Age`.

The latest year available in the dataset is used as the reference year.

```
In [34]: latest_year = df["Year"].max()

df["Car_Age"] = latest_year - df["Year"]

print("Reference year:", latest_year)

df[["Year", "Car_Age"]].head()
```

Reference year: 2018

```
Out[34]:
```

	Year	Car_Age
0	2014	4
1	2013	5
2	2017	1
3	2011	7
4	2014	4

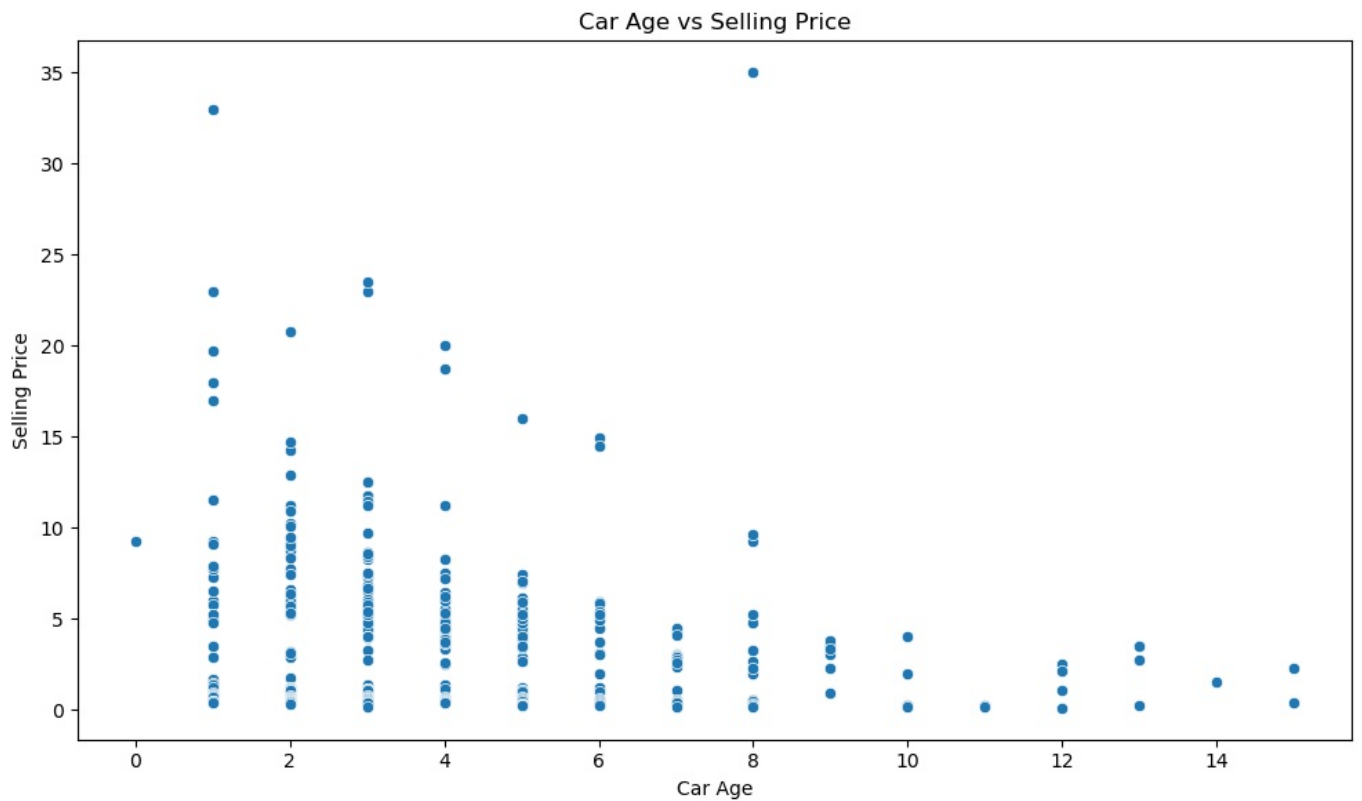
18. Car Age vs Selling Price

Car age is compared with selling price to understand whether older vehicles generally have lower resale prices.

```
In [48]: plt.figure(figsize=(10, 6))

sns.scatterplot(
    data=df,
    x="Car_Age",
    y="Selling_Price"
)
```

```
plt.title("Car Age vs Selling Price")
plt.xlabel("Car Age")
plt.ylabel("Selling Price")
plt.tight_layout()
plt.show()
```



19. Feature Correlation Heatmap

A correlation heatmap is used to examine the relationships between the numerical variables in the dataset.

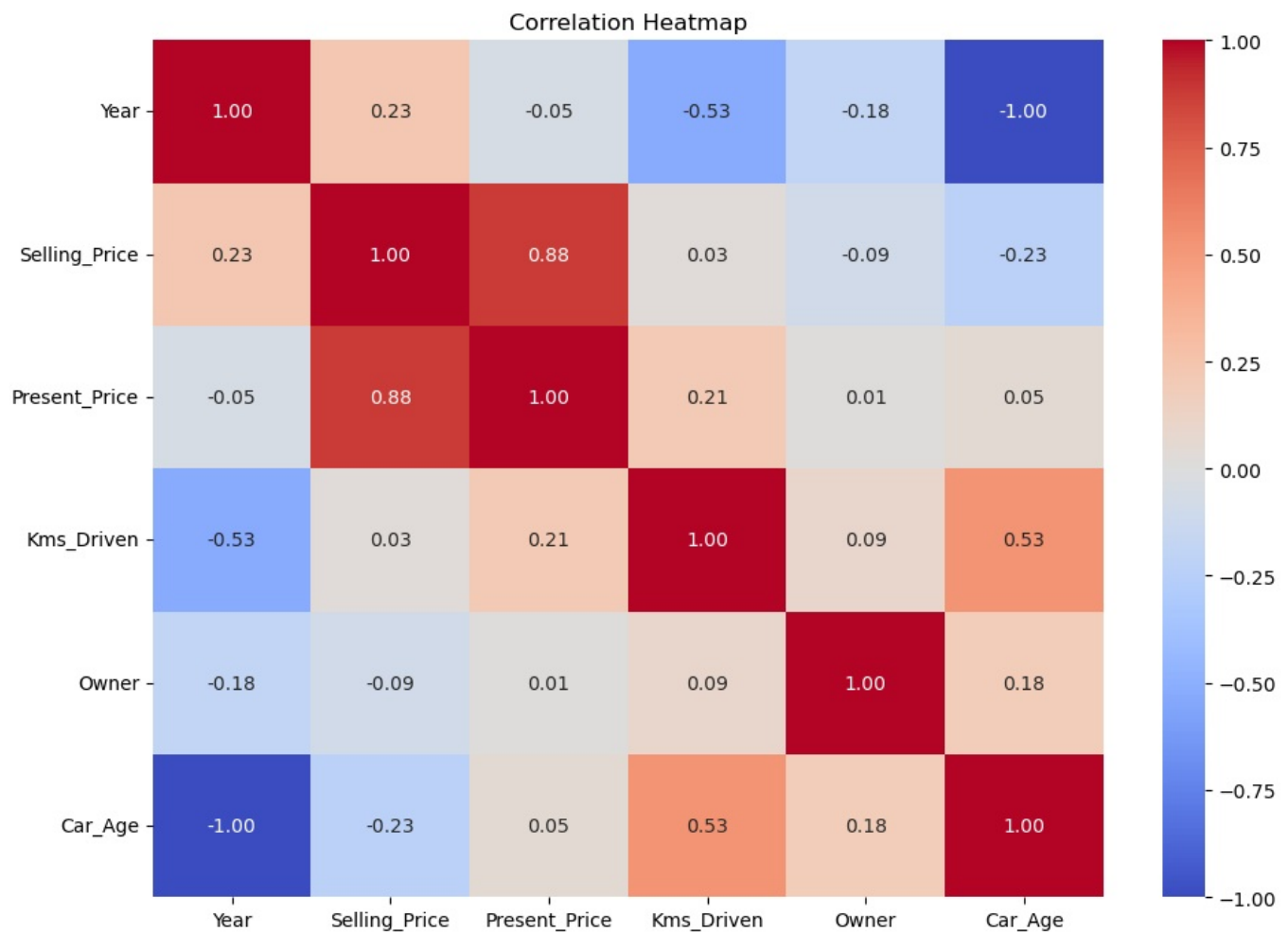
The heatmap helps identify which numerical features have stronger relationships with the selling price.

```
In [49]: plt.figure(figsize=(10, 7))

correlation = df[
    [
        "Year",
        "Selling_Price",
        "Present_Price",
        "Kms_Driven",
        "Owner",
        "Car_Age"
    ]
].corr()

sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Heatmap")
plt.tight_layout()
plt.show()
```



Observation

The correlation heatmap shows the strength and direction of relationships between the numerical variables. Present price has a strong positive relationship with selling price, while car age and kilometres driven show negative relationships with selling price.

20. Feature Engineering — Car Brand

The car name is used to create a new feature called `Car_Brand`.

The first part of the car name is extracted as a simple brand or model prefix. This gives the model additional information about the type of vehicle.

```
In [46]: df["Car_Brand"] = df["Car_Name"].str.split().str[0].str.title()
print("Car brand feature created successfully.")
df[["Car_Name", "Car_Brand"]].head(10)
```

Car brand feature created successfully.

```
Out[46]:
```

	Car_Name	Car_Brand
0	ritz	Ritz
1	sx4	Sx4
2	ciaz	Ciaz
3	wagon r	Wagon
4	swift	Swift
5	vitara brezza	Vitara
6	ciaz	Ciaz
7	s cross	S
8	ciaz	Ciaz
9	ciaz	Ciaz

21. Feature and Target Selection

The target variable is `Selling_Price`.

The remaining selected variables will be used as input features for the machine learning models.

```
In [50]: features = [
    "Present_Price",
    "Kms_Driven",
    "Fuel_Type",
    "Seller_Type",
    "Transmission",
    "Owner",
    "Car_Age",
    "Car_Brand"]

X = df[features]
y = df["Selling_Price"]

print("Features:")
print(features)

print("\nTarget:")
print("Selling_Price")
```

Features:

['Present_Price', 'Kms_Driven', 'Fuel_Type', 'Seller_Type', 'Transmission', 'Owner', 'Car_Age', 'Car_Brand']

Target:

Selling_Price

22. Train-Test Split

The dataset is divided into training and testing data.

80% of the observations are used for training and 20% are used for testing the machine learning models.

```
In [51]: X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)

print("Training records:", len(X_train))
print("Testing records:", len(X_test))
```

Training records: 239

Testing records: 60

23. Data Preprocessing

Categorical variables are converted into numerical values using One-Hot Encoding.

```
In [59]: categorical_features = [
    "Fuel_Type",
    "Seller_Type",
    "Transmission",
    "Car_Brand"]

numerical_features = [
    "Present_Price",
    "Kms_Driven",
    "Owner",
    "Car_Age"]

preprocessor = ColumnTransformer(
    transformers=[
        (
            "categorical",
            OneHotEncoder(handle_unknown="ignore"),
            categorical_features
        ),
    ],
    remainder="passthrough"
)

print("Preprocessing pipeline updated successfully.")
```

Preprocessing pipeline updated successfully.

24. Linear Regression

Linear Regression is used as a baseline regression model for predicting car selling prices.

```
In [60]: linear_model = Pipeline([
          ("preprocessor", preprocessor),
          ("model", LinearRegression())])

linear_model.fit(X_train, y_train)

linear_predictions = linear_model.predict(X_test)

print("Linear Regression model trained successfully.")
```

Linear Regression model trained successfully.

25. Decision Tree Regression

A Decision Tree Regressor is used to model non-linear relationships between the car features and selling price.

```
In [61]: tree_model = Pipeline([
          ("preprocessor", preprocessor),
          ("model", DecisionTreeRegressor(max_depth=8, random_state=42))])

tree_model.fit(X_train, y_train)

tree_predictions = tree_model.predict(X_test)

print("Decision Tree model trained successfully.")
```

Decision Tree model trained successfully.

26. Random Forest Regression

A Random Forest Regressor is trained using multiple decision trees. This model is used to improve prediction performance and reduce the effect of an individual tree.

```
In [62]: forest_model = Pipeline([
          ("preprocessor", preprocessor),
          ("model", RandomForestRegressor(n_estimators=200, max_depth=10, random_state=42))])

forest_model.fit(X_train, y_train)

forest_predictions = forest_model.predict(X_test)

print("Random Forest model trained successfully.")
```

Random Forest model trained successfully.

27. Model Evaluation

The models are evaluated using:

- MAE — Mean Absolute Error
- MSE — Mean Squared Error
- RMSE — Root Mean Squared Error
- R² — Coefficient of Determination

Lower MAE, MSE and RMSE indicate lower prediction error. A higher R² indicates better explanatory performance.

```
In [64]: def evaluate_model(name, actual, predicted):

          mse = mean_squared_error(actual, predicted)

          return {"Model": name, "MAE": mean_absolute_error(actual, predicted), "MSE": mse, "RMSE": np.sqrt(mse), "R2": r2_score(actual, predicted)}

results_df = pd.DataFrame([evaluate_model("Linear Regression", y_test, linear_predictions),
                             evaluate_model("Decision Tree", y_test, tree_predictions),
                             evaluate_model("Random Forest", y_test, forest_predictions)])

results_df
```

Out[64]:	Model	MAE	MSE	RMSE	R2
0	Linear Regression	1.348138	5.707221	2.388979	0.778560
1	Decision Tree	0.999118	3.928858	1.982135	0.847561
2	Random Forest	1.420197	11.989887	3.462642	0.534794

28. Best Performing Model

The model with the highest R^2 score is selected as the best-performing model for this analysis.

```
In [65]: best_model_name = results_df.loc[
        results_df["R2"].idxmax(),
        "Model"]

if best_model_name == "Linear Regression":
    best_model = linear_model
    best_predictions = linear_predictions

elif best_model_name == "Decision Tree":
    best_model = tree_model
    best_predictions = tree_predictions

else:
    best_model = forest_model
    best_predictions = forest_predictions

print("Best model:", best_model_name)
```

Best model: Decision Tree

29. Feature Importance

Feature importance is used to understand which features contribute most to the predictions made by the best-performing tree-based model.

This helps identify the factors that are most useful when estimating the selling price of used cars.

```
In [67]: if best_model_name in ["Decision Tree", "Random Forest"]:

    model = best_model.named_steps["model"]
    fitted_preprocessor = best_model.named_steps["preprocessor"]

    feature_names = fitted_preprocessor.get_feature_names_out()
    importances = model.feature_importances_

    importance_df = pd.DataFrame({
        "Feature": feature_names,
        "Importance": importances
    })

    importance_df = importance_df.sort_values(
        "Importance",
        ascending=False
    ).head(10)

    print("Top 10 important features:")
    display(importance_df)

    plt.figure(figsize=(10, 6))

    sns.barplot(
        data=importance_df,
        x="Importance",
        y="Feature"
    )

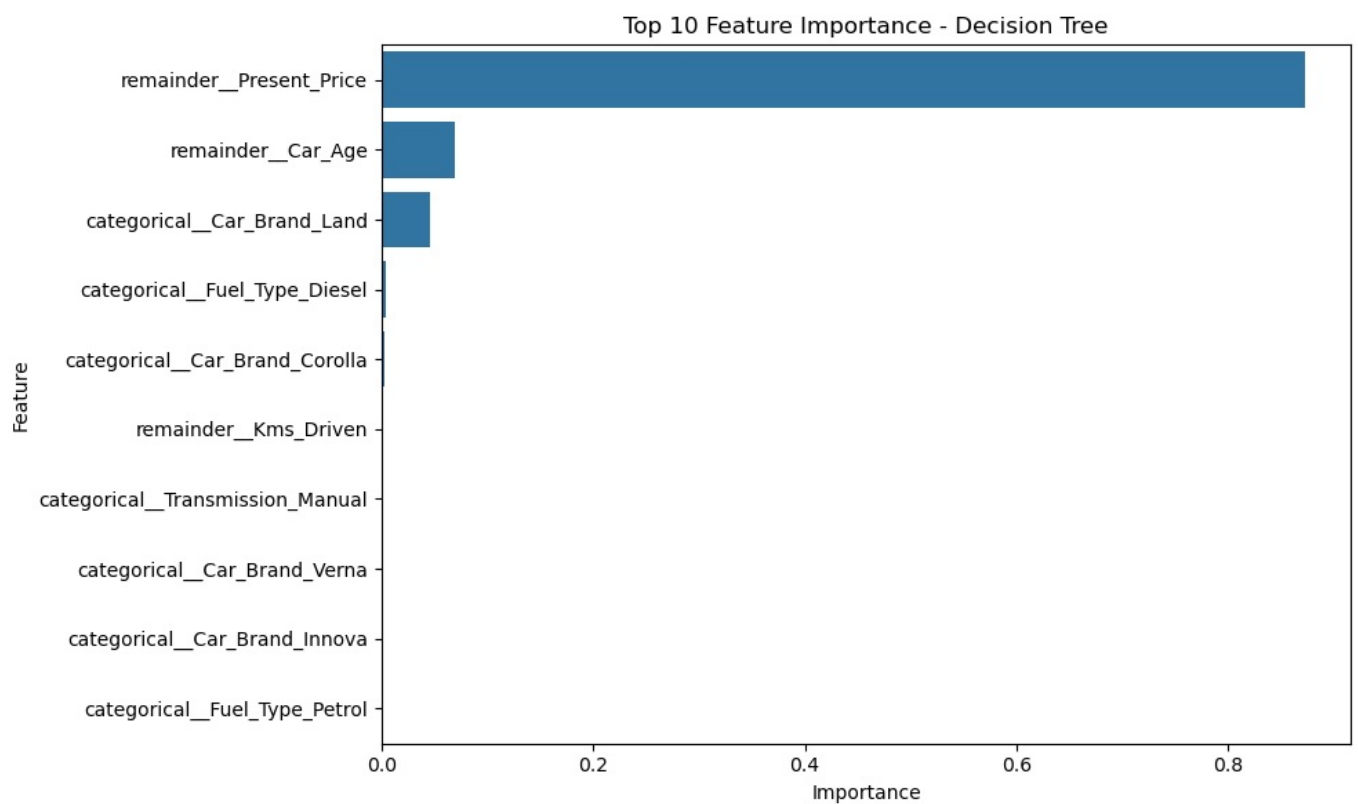
    plt.title(f"Top 10 Feature Importance - {best_model_name}")
    plt.xlabel("Importance")
    plt.ylabel("Feature")

    plt.tight_layout()
    plt.show()

else:
    print("Feature importance is not available for Linear Regression.")
```

Top 10 important features:

	Feature	Importance
46	remainder__Present_Price	0.872172
49	remainder__Car_Age	0.069445
33	categorical__Car_Brand_Land	0.045938
1	categorical__Fuel_Type_Diesel	0.004294
16	categorical__Car_Brand_Corolla	0.002663
47	remainder__Kms_Driven	0.002050
6	categorical__Transmission_Manual	0.001192
42	categorical__Car_Brand_Verna	0.000760
30	categorical__Car_Brand_Innova	0.000340
2	categorical__Fuel_Type_Petrol	0.000326



30. Actual vs Predicted Selling Price

The actual selling prices are compared with the predictions generated by the best-performing model.

```
In [45]: plt.figure(figsize=(8, 6))
plt.scatter(y_test,best_predictions)

minimum = min(y_test.min(),best_predictions.min())
maximum = max(y_test.max(),best_predictions.max())

plt.plot([minimum, maximum],[minimum, maximum],linestyle="--")

plt.title(f"Actual vs Predicted Selling Price - {best_model_name}")

plt.xlabel("Actual Selling Price")
plt.ylabel("Predicted Selling Price")

plt.tight_layout()
plt.show()
```



31. Key Findings

Based on the analysis and model results, the following findings were observed:

1. Present price shows a clear positive relationship with selling price. Cars with higher present prices generally have higher selling prices.
2. Newer cars generally tend to have higher selling prices, although the selling price varies between individual vehicles.
3. Cars with higher kilometers driven generally tend to have lower selling prices, although this relationship is not consistent for every vehicle.
4. Diesel cars have the highest average selling price among the fuel types in this dataset, with an average of approximately 10.10.
5. Automatic cars have a higher average selling price than manual cars, with average selling prices of approximately 9.07 and 3.92 respectively.
6. Among the three machine learning models tested, Decision Tree Regression achieved the best overall performance.
7. The Decision Tree model achieved an MAE of 0.999, an RMSE of 1.982, and an R^2 score of 0.848.
8. Present Price was the most important feature in the Decision Tree model, followed by Car Age and some car brand/model categories.
9. Overall, the results show that present price, car age, kilometers driven, fuel type, seller type, transmission, ownership, and car brand/model information can be useful for predicting used-car selling prices.

32. Conclusion

This project used Python and machine learning to predict the selling price of used cars.

The analysis included data cleaning, exploratory data analysis, feature engineering, categorical encoding, train-test splitting, regression modelling, and model evaluation.

Three models were tested: Linear Regression, Decision Tree Regression, and Random Forest Regression. Based on the test results, Decision Tree Regression performed the best, achieving an R^2 score of approximately 0.848, along with an MAE of approximately 0.999 and an RMSE of approximately 1.982.

The feature importance analysis showed that Present Price was the most important feature for the Decision Tree model, followed by Car Age and selected car brand/model categories.

Overall, the project demonstrates that machine learning can be used to estimate used-car selling prices from vehicle-related information. However, actual selling prices may also depend on other factors such as vehicle condition, service history, location, and market demand, which are not included in this dataset.

In []: